LEE HENG WAI ALAN | (+65) 91510925 | ahlanlhw@gmail.com | Singapore | LinkedIn | GitHub | Writing

CAREER SUMMARY

Data analyst and engineer with 10+ years across e-commerce, logistics, energy, finance and crypto. At Shopee, built production pipelines in Python, SQL and PySpark, ran A/B tests on the Shopee Express locker rollout and expansion, and built the backlog dashboards logistics leadership used to monitor operations. Most recently built sprkly.app, a social analytics SaaS that brings engagement data from five platform APIs into one schema.

PROFESSIONAL EXPERIENCE

sprkly.app, SINGAPORE
Founder & Engineer, sprkly.app **Feb 2026 – Present**

- Built the data layer for sprkly.app: scheduled ingestion from the Facebook, Instagram, Threads, YouTube and TikTok APIs, normalised into a common schema that powers cross-platform analytics and posting-time recommendations for creators, social media managers and agencies.
- Shipped an MCP server so customers can query their own analytics and schedule posts through any AI assistant, without opening a dashboard.
- Cleared production app review with Meta, Google and TikTok, covering business verification, scope justification and data-handling compliance.

SHOPEE SINGAPORE, SEA GROUP, SINGAPORE
Senior BI Analyst (ML & Automation Engineering) **Feb 2024 – Jun 2026**

- Maintained and tuned the in-house XGBoost model for Shopee SG's Flash Deals SKU selection, covering ~10% of platform daily GMV. Owned end-to-end preprocessing and feature pipelines in Python and SQL.
- Built production pipelines orchestrated in Airflow and later Shopee's in-house scheduler. Used PySpark for heavy batch jobs and PrestoSQL for most analysis, where fast turnaround mattered more than scale.
- Ran A/B tests during the Shopee Express locker rollout comparing locker vs doorstep delivery. Over 1 in 3 users in test areas adopted lockers, with the strongest uptake in buyer-dense areas, lowering doorstep delivery costs.
- Built logistics dashboards used directly by leadership to spot and act on delivery backlogs.
- Built agentic Python workflows automating image audits at scale, with the agent calling Gemini Flash and Claude Sonnet for audit decisions. Replaced manual review and delivered annualised six-figure cost savings.

WEMIX SINGAPORE, WEMADE KOREA
Business & Portfolio Investments Analyst **May 2023 – Feb 2024**

- Led India market expansion research; delivered on-chain analysis and portfolio company support to C-suite.

CRYPTO.COM, SINGAPORE
Research Analyst, Blockchain & DeFi **Feb 2022 – Aug 2022**

- Built a daily RSS aggregator automating research collection via the Google Developer API. Wrote monthly technical articles on blockchain infrastructure.

Wood Mackenzie, VERISK Analytics
Data Analyst, Upstream Oil & Gas **Jan 2019 – Feb 2022**

- Owned the North and Southeast Asia upstream data offering. Ran ETL ingestion and automation on AWS S3 and Glue, and governed production datasets delivered to clients.

Earlier: *Research Analyst*, **J.P. Morgan Chase** (EM Asia Economics, **2018**) · *Research Analyst*, **CAMRI, NUS Business School** (quantitative modelling and portfolio optimisation, **2015–2018**)

PROJECTS

- Skillhack.io ***(launched Aug 2026)*** **—** co-built an AI skills assessment platform, with prompts scored by an LLM judge against a five-criteria rubric.
- **Low-Latency Market-Making System (C#) —** event-driven order handling, volatility estimation and queue-position logic, deployed on a Linux VPS. Shut down when measured alpha turned negative.

TECHNICAL SKILLS

- **ML-related**: LLM-as-judge evaluation, prompt design, XGBoost, Scikit-learn, NumPy, pandas, feature engineering
- **Data & Infra**: PostgreSQL, D1/SQLite, NoSQL, Presto, Airflow, dbt, Databricks, pySpark
- **AI Tools**: Claude Code, Codex, MCP Server design
- **Engineering**: Python, SQL, TypeScript, C#, Cloudflare Workers, AWS, GCP, Docker, Git
- **Languages**: English, Mandarin, Japanese (basic, N4 equivalent)

EDUCATION

UNIVERSITY OF LONDON INTERNATIONAL PROGRAMMES, BSC ECONOMICS, UPPER SECOND CLASS HONOURS 2012 – 2015